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**BREATHETWIN**

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*EARLY WARNING. EASIER BREATHING.*

# BreatheTwin

## *A COPD digital twin for India*

Warns 1 to 3 days before a flare-up, explains why, and shows what to do


**Team MOMENTUM · SDA Bocconi Asia Center, Mumbai**

*Digital Twin Challenge 2026 · Happiest Health*

**AT**

Anushka Tomar

*Team leader*

**YS**

Yash Saxena

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# Flare-ups warn days ahead, yet India catches them too late

### India's COPD burden is large

- **55 million** people in India lived with COPD in 2016 *(India State-Level Disease Burden Initiative, Lancet Global Health 2018)*
- Chronic respiratory disease caused **10.9%** of all deaths in India in 2016 *(same study)*

### Flare-ups are the dangerous moments

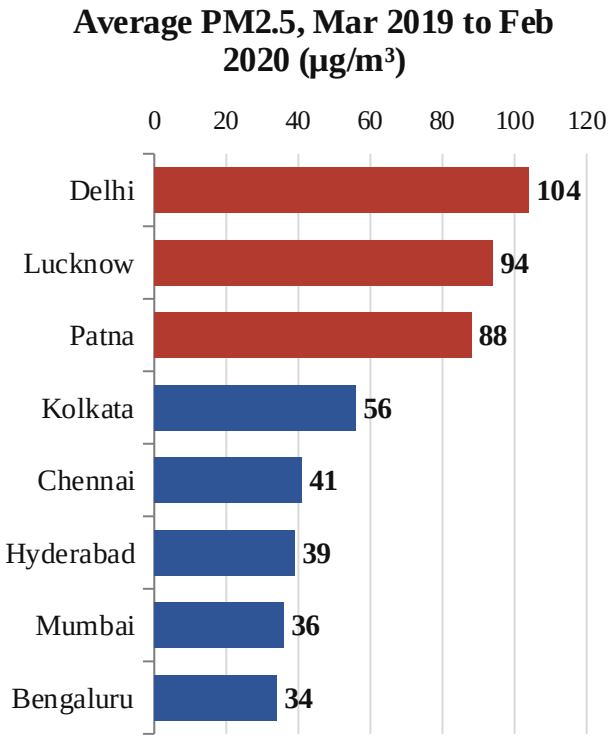
- Patients have about **1 to 2 flare-ups a year**, more as COPD worsens *(ECLIPSE, NEJM 2010)*
- Warning signs start **5 to 7 days** earlier: breathing rises by **4.4 breaths/min** the day before *(Yañez 2012; Hurst 2010)*
- Today they are noticed only when the patient is already struggling

## The Care Gap

| Care today | What patients need |
|---|---|
| Check-ups every few months | A daily watch against their own normal |
| Acts after symptoms appear | A warning 1 to 3 days ahead |
| Same alarm limits for everyone | Limits personal to each patient |
| Ignores city air and weather | Record, body and air in one view |

**Air makes it worse *(real CPCB data, 2019-20)***

- Delhi PM2.5 averaged **104 µg/m³**, about **7×** the WHO daily safe level of 15
- Winter air was **3× dirtier** than monsoon air (104 vs 32 µg/m³)




## The opportunity:

*Use the days of warning the body already gives: a personal twin that spots a flare-up early, explains why, and tells families what to do.*

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# BreatheTwin: a twin that warns 1 to 3 days before a flare-up

## Core Proposition

**A personal twin for every COPD patient that:**

- Learns each person's **own normal**
- Predicts a **flare-up 1 to 3 days ahead**
- **Explains the reasons** in plain words
- Shows what would **lower the risk**
- Gives families **smog warnings** and a **visit summary** for the doctor

## 3 Data Streams

| Stream | What it adds |
|---|---|
| **Health record (EHR)** | Severity, past flare-ups, smoking, blood tests, other illnesses: the *starting risk* |
| **Wearable + smart inhaler** | Daily heart rate, breathing, oxygen, steps, sleep, inhaler use: the *early warning* |
| **City air + weather** | PM2.5, cold snaps, humidity: the *seasonal trigger* and the advice |

## Tech & Experience




*Doctor's view: the risk, who to call today, and why*

## Key Differentiator

| Usual care | Wearables | BreatheTwin |
|---|---|---|
| Reactive | Raw numbers | **Predictive** |
| Textbook limits | Textbook limits | **Own normal** |
| No air data | No record | **Record + body + air** |
| Doctor only | User only | **Doctor + family** |

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# Works with the wearables and IoT devices patients can already buy

## Which sensor tracks which signal

| Signal | Sensor | Devices | Twin uses it as |
|---|---|---|---|
| Resting heart rate | Optical PPG (green light) | Any smartwatch, band, oximeter | Change vs own normal |
| Breathing rate | PPG + motion, during sleep | Fitbit, Apple, Galaxy, Garmin watch | Change vs own normal |
| Blood oxygen (SpO2) | Red + infrared light | Fingertip / ring oximeter, watch | Change; urgent below 90% |
| Steps / activity | Accelerometer | Watch, band or phone | Change vs own normal |
| Sleep hours | Accelerometer + heart rate | Watch or band | Change vs own normal |
| Rescue puffs, doses | Clip-on actuation sensor | ADHERO (Lupin), Propeller, Hailie | Puffs vs normal; adherence |
| Blood pressure | Cuff pressure (oscillometric) | Omron Bluetooth monitor | Shown to the doctor |
| PM2.5, temperature | CPCB stations, weather API | City networks (no device) | Dose-response, cold snaps |

## Three ways to equip a patient

**Minimum kit:** Any watch that syncs to Health Connect or Apple Health + fingertip oximeter. Gives heart rate, breathing, SpO2, steps, sleep.

**Full kit:** + smart inhaler clip (ADHERO, launched in India by Lupin) + Omron BP cuff. Adds automatic inhaler tracking.

**No-watch kit:** Phone + oximeter; the family types SpO2 and a 30-second breathing count. Near-zero cost.

**No needle, no blood test, nothing under the skin:**

*a watch and a clip-on oximeter carry the daily signal; data flows phone → Health Connect / HealthKit → BreatheTwin.*

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## How the twin works: three data streams, one personal model

### Data In

**Health record (EHR)**
*420 synthetic patients: Synthea + India overlay*

**Wearable + smart inhaler**
*heart rate, breathing, SpO2, steps, sleep, puffs, doses*

**City air + weather**
*real CPCB PM2.5 and Open-Meteo, 8 cities*

### The Twin

**1. Personal baseline**
Every reading compared with the patient's **own last 4 weeks**, not a textbook range

**2. Hybrid risk engine**
**LightGBM** learns body signals, inhaler use, weather and record × **published PM2.5 effect** (DeVries 2017)

### What Comes Out

- **Risk of a flare-up in the next 3 days**
- **Reasons, ranked by how much they push risk**
- **Safety checks against own normal**
- **What-if answers for doctor and family**
- **Smog warnings + one-page visit summary**

### Twin journey

**Health record** → **Daily readings** → **Own normal** → **Risk score** → **Reasons** → **Action + advice**

*Why hybrid: the body-signal patterns are complex and best learned from data; the air-pollution effect is small and well measured in large studies, so we take it from the literature.*

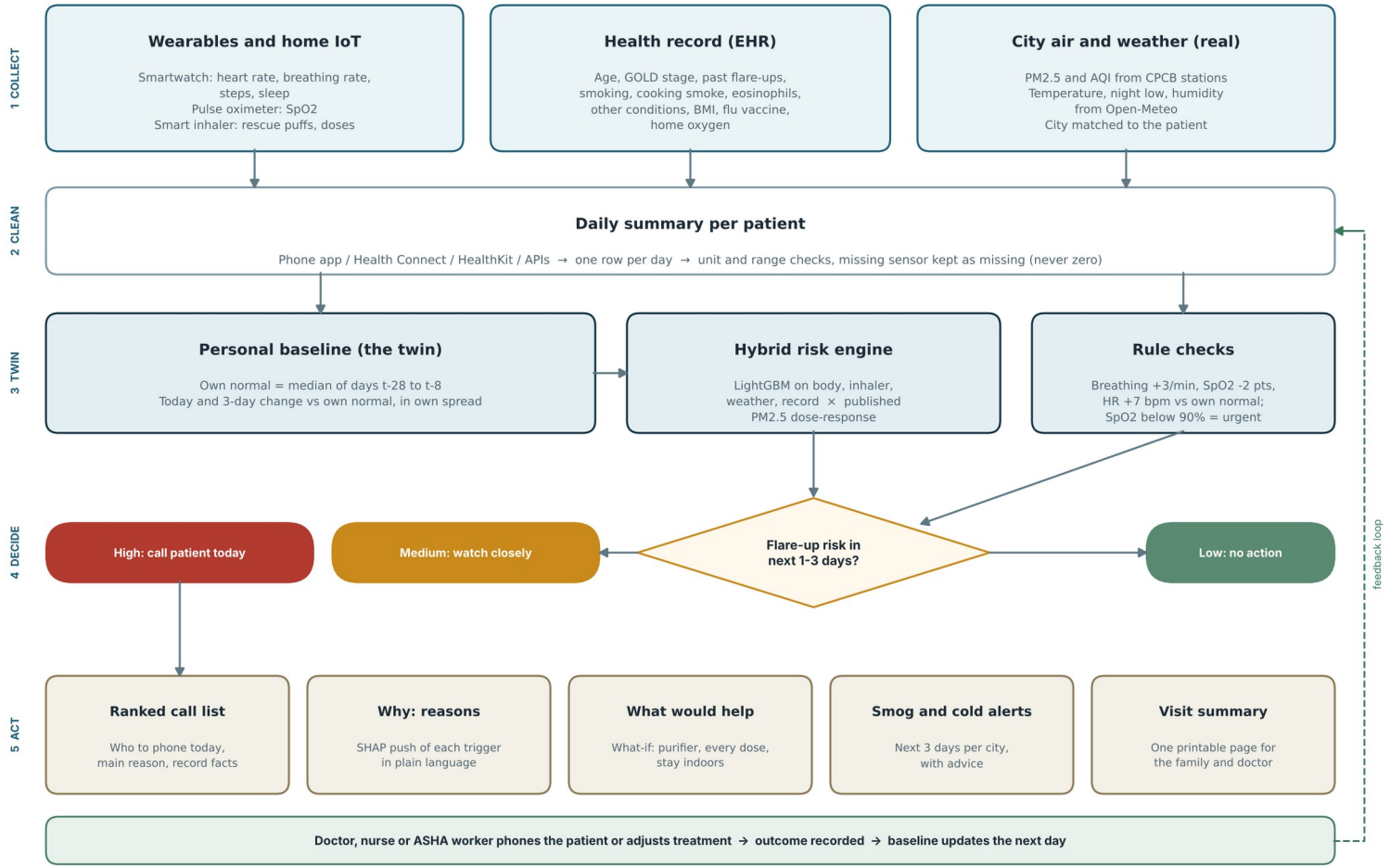
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# Flowchart: what BreatheTwin does for every patient, every morning




**1 Collect.** Watch, oximeter, inhaler clip; health record; real city air and weather.

**2 Clean.** One row per patient per day; range checks; missing stays missing.

**3 Twin.** Today vs own normal (days t-28 to t-8), hybrid risk engine, rule checks.

**4 Decide.** High: call today. Medium: watch. Low: no action.

**5 Act.** Call list, reasons, what-if, smog alerts, visit summary; outcome feeds back.

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# Open, anonymised or synthetic data, and clear about which

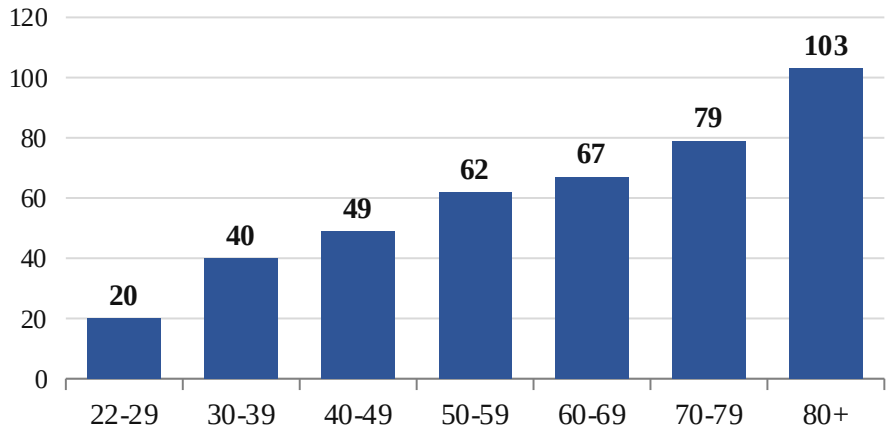
## Data Sources

| Stream | Source | Real or synthetic | Licence |
|---|---|---|---|
| Health records | Synthea COPD cohort + our India overlay and younger patients | Synthetic | Apache 2.0 |
| Wearables and flare-ups | Simulated; every effect size from a published study | Synthetic | Ours (MIT) |
| Air quality | CPCB daily PM2.5, 8 cities (Kaggle 'Air Quality Data in India') | **Real** | CC0 |
| Weather | Open-Meteo archive: temperature, humidity | **Real** | CC BY 4.0 |
| Real-data check | MIMIC-IV Clinical Database Demo v2.2 (14 COPD patients) | **Real**, aggregate only | ODbL 1.0 |

**Real-data check:** sex mix and heart disease match real patients; heart failure and diabetes are lower in synthetic records.

## Our Patient Cohort (420)

**Patients by age group: every adult generation**




- Severity GOLD 1 / 2 / 3 / 4: **18% / 43% / 28% / 11%**
- Current smokers 19% · cooking smoke at home 32% · 8 cities

## Honest by design:

*Air and weather are real; patients, watch readings and flare-ups are simulated, so the results prove the idea works, not clinical accuracy.*

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# How much of our data is real, and what each part is used for

## Every data stream, real or synthetic

| Data | Real? | Size | Used for |
|---|---|---|---|
| City air quality (PM2.5, AQI) | Real | 2,928 city-days, 8 cities (CPCB) | Training + dashboard |
| Weather (temperature, humidity) | Real | 2,928 city-days (Open-Meteo) | Training + dashboard |
| Health records | Synthetic | 420 patients (Synthea + India overlay, 20 under 30) | Training + dashboard |
| Watch, oximeter, inhaler readings | Synthetic | 153,720 patient-days, from studies | Training + dashboard |
| Flare-ups | Synthetic | 454 simulated events | Training labels |
| Real hospital patients (stress test) | Real | 100 patients, 47,214 vital readings | Testing only |
| Real COPD records (check) | Real | 14 COPD patients (MIMIC-IV demo) | Testing only |

**100%**

**of air and weather data**

*is real (CPCB, Open-Meteo)*

**~1%**

**of all training values**

*are real; patient data is simulated*

**100**

**real patients, 55 women, 45 men**

*aged 18 to 90, used to stress-test*

## Why simulate patients at all?

*No open dataset links COPD wearable data to labelled flare-ups, so we simulate from published studies and test on real people.*

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# AI model: learn the body, borrow published science for the air

## What the Model Looks At

| Data input | Signal the twin uses |
|---|---|
| Breathing, oxygen, heart rate | Change vs **own normal** (today, 3-day average, how unusual) |
| Steps and sleep | Drop vs own normal |
| Inhalers | Extra rescue puffs; % of daily doses taken this week |
| Weather | 3-week temperature drop, night low, humid heat |
| Health record | Age, GOLD stage, past flare-ups, smoking, cooking smoke, eosinophils, other illnesses, BMI, flu vaccine, home oxygen |
| Air (PM2.5) | Published dose-response, **not learned**: +2.5% risk per 10 µg/m³ |

**Tech stack:** Python · pandas · LightGBM · scikit-learn · Streamlit · Plotly · Synthea (Java)

**Open source:** MIT licence; whole pipeline rebuilds with one command

## Model Details

| | |
|---|---|
| **Algorithm** | LightGBM (gradient-boosted decision trees) |
| **Predicts** | Will a flare-up start in the next 1 to 3 days? |
| **Tested on** | 126 patients never seen in training (294 trained) |
| **Safety rules** | Warning signs can never lower risk; more inhaler doses can never raise it |
| **Explanations** | Per-factor contributions (SHAP values): "what is pushing the risk" |

### Every number has a source

- Breathing rise before flare-ups: *Yañez 2012*
- Flare-up rates by severity: *ECLIPSE, Hurst 2010*
- Eosinophils ≥340/µL: **1.76×** flare-ups *(Vedel-Krogh 2016)*
- Still smoking: **1.28×** *(Au 2009)* · PM2.5: *DeVries 2017*
- Anything unsourced is labelled **ASSUMPTION** in the code

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## Results: 2 in 3 flare-ups flagged 1 to 3 days ahead

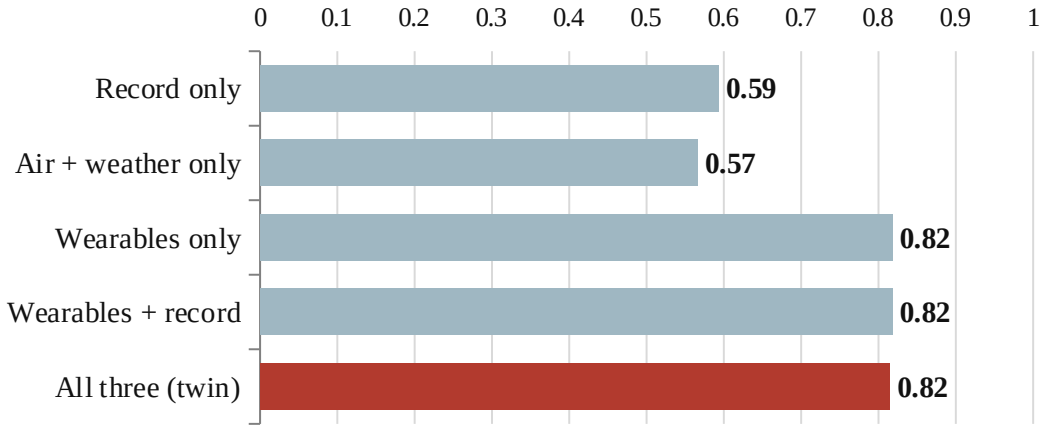
**0.815**
**Accuracy score (AUROC)**
*published tools: 0.68 to 0.85*

**67%**
**Flare-ups warned**
*1 to 3 days ahead*

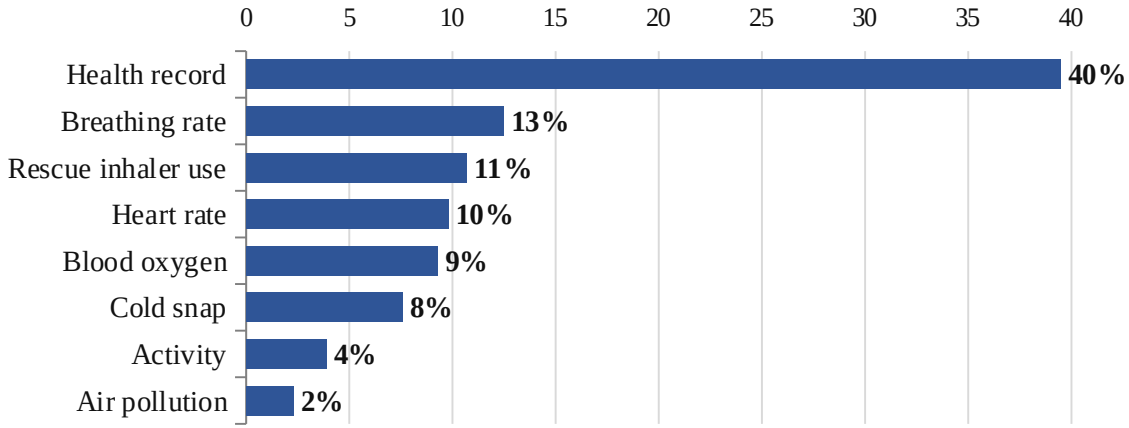
**0.91**
**False alerts**
*per patient per month*

**126**
**Unseen test patients**
*never used in training*

### Prediction quality by data stream (AUROC)




### Share of the twin's reasoning by factor (%)




### What this means:

*Body signals carry the early warning; the health record sets each person's starting risk; air and weather explain the seasons.*

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## Stress test on 100 real patients: robust, fair, right direction

**75/100**
**real patients assessed**
*25 had under 48 h of vitals*

**0**
**crashes or invalid risks**
*incl. missing oxygen sensor*

**0.73**
**risk vs real worsening**
*rank correlation*

**Pass**
**worse readings raise risk**
*8-case edge suite*

### By sex and age group

| Group | Patients | Median risk | Alerted |
|---|---|---|---|
| Female | 44 | 0.9% | 30% |
| Male | 31 | 0.6% | 23% |
| 18-39 | 2 | 15.6% | 50% |
| 40-49 | 6 | 0.9% | 33% |
| 50-59 | 7 | 0.9% | 29% |
| 60-69 | 9 | 0.6% | 11% |
| 70-79 | 18 | 0.5% | 17% |
| 80+ | 32 | 0.8% | 34% |

### What we learned

**Robust:** every real patient with 48 h of data was scored; no crash, no impossible risk.

**Right direction:** breathing up to 35/min gives High risk; all three signs worse gives 97%.

**Age and sex:** age 18 vs 90 and switching sex barely change risk; sex is not a model input.

**Fixed:** oxygen at 82% scored Low, so a safety override now makes any reading below 90% Urgent.

**Limits:** no link with later in-hospital death (AUROC 0.51); women alerted more at equal signs, but groups are 5 to 19 people.

**Real people, never used for training:**
*MIMIC-III demo v1.4 (PhysioNet, ODbL). ICU patients are not a COPD home cohort: a stress test, not validation.*

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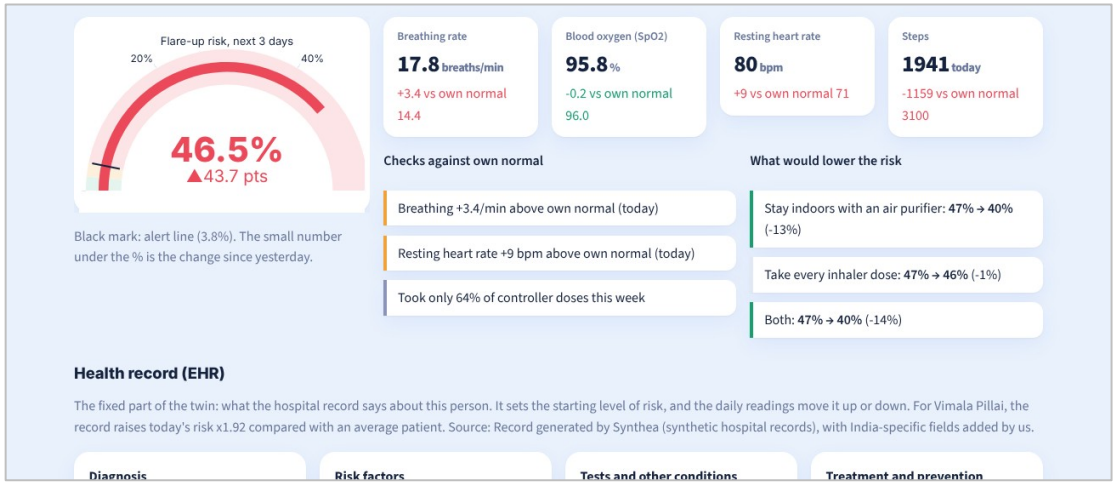
# The doctor's dashboard: who to call today, and why

## Overview: all 420 patients




- **Call these patients today**: ranked by risk, with the main reason and key record facts
- India map of risk and air; filter by age group; plain-language guide on every page

## One patient in detail




- Risk dial with the alert line and the change since yesterday
- Today's readings vs **own normal**: red means worse than usual
- **What would lower the risk**: e.g. air purifier 49% → 43%

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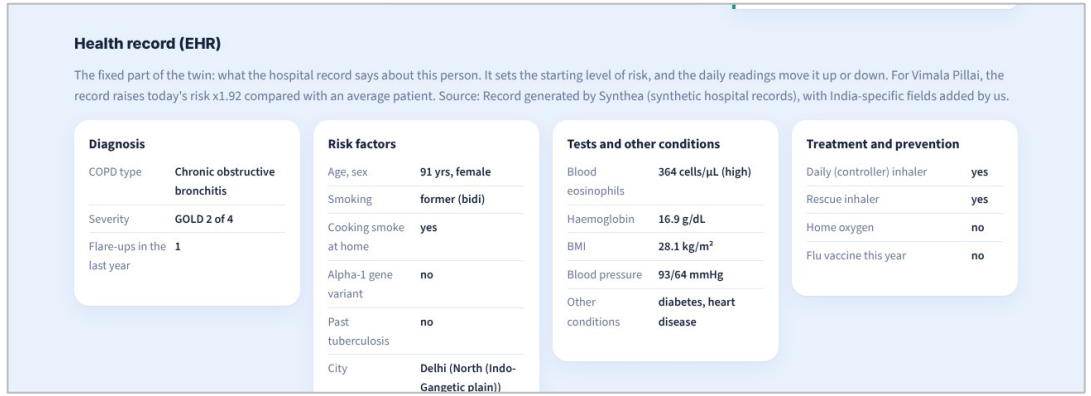
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# Health record in every view, and a lab to test advice before giving it

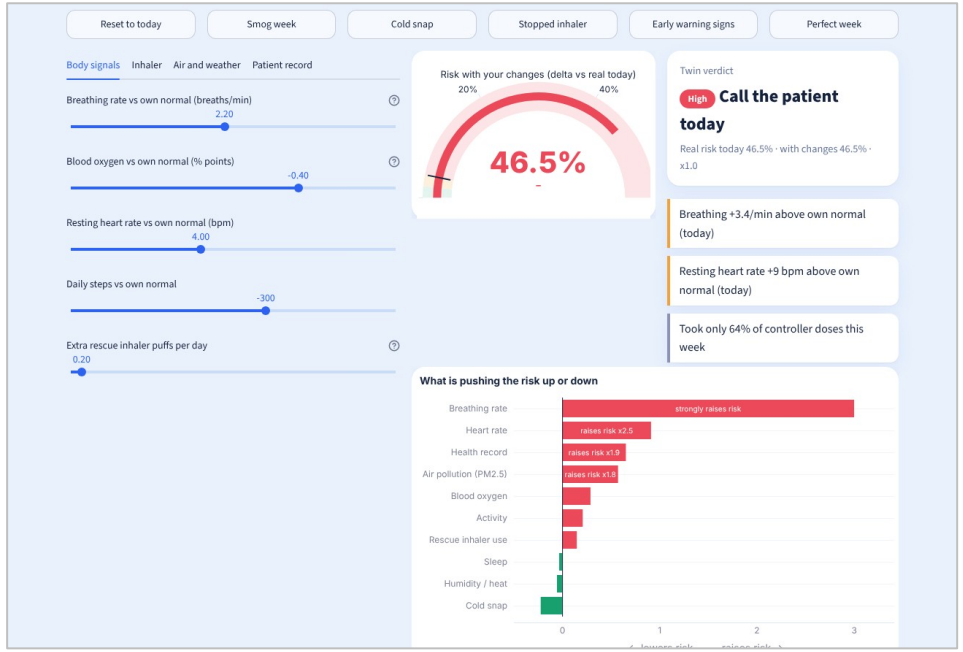
## Health Record (EHR)




**The fixed part of the twin**

- Diagnosis, risk factors, blood tests, other illnesses, medicines
- Shows how much the record **raises or lowers** today's risk
- Record facts also appear on the call list, what-if lab and new-person check

## What-if Lab




- One click scenarios: **smog week, cold snap, stopped inhaler, early warning signs, perfect week**
- Sliders for body signals, inhaler, air, weather and the record; risk re-scores instantly

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# Check a new person in minutes, and see every value the twin used

## Result for a new person with warning signs




## How it works

1. Health record: age, severity, smoking, blood tests
2. Readings **usual** (a good day) and **now** (last 3 days)
3. Optional: today's PM2.5 from a CPCB or AQI app
4. Risk, reasons, safety checks and a table of every value used

## We tested it

| Test case | Result |
|---|---|
| Readings now = usual | **Low, about 0.3%** |
| Breathing 16 → 22, oxygen 95 → 91%, puffs 1 → 4 | **High, about 99%** |
| Eosinophils 220 → 520 cells/µL | Risk 0.26% → 0.33% |

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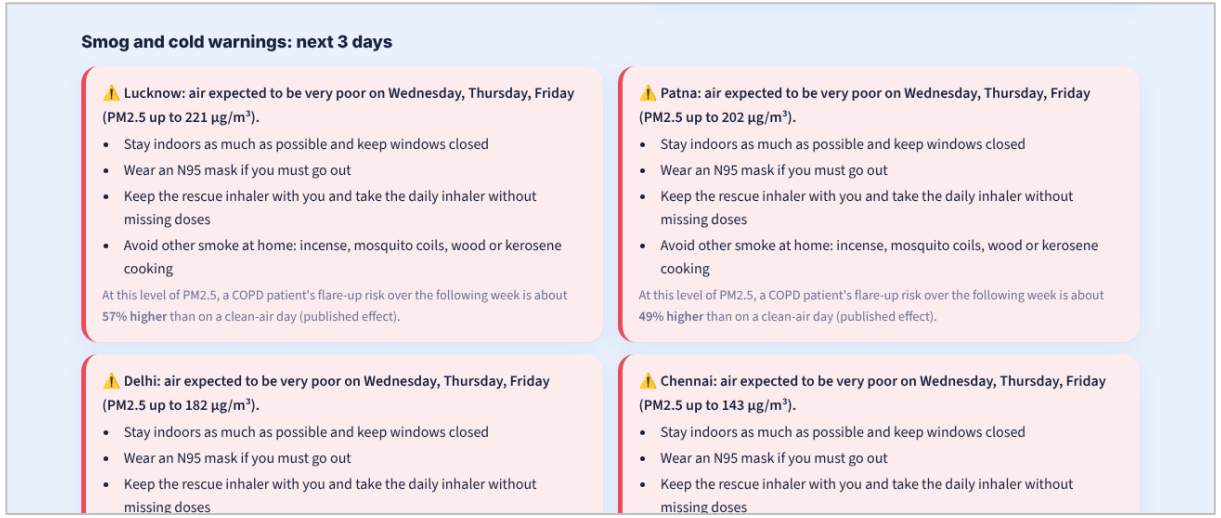
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# From prediction to action: smog warnings and a visit summary

## Smog is Coming: next 3 days



**Smog and cold warnings: next 3 days**

⚠️ **Lucknow: air expected to be very poor on Wednesday, Thursday, Friday (PM2.5 up to 221 µg/m³).**

- Stay indoors as much as possible and keep windows closed
- Wear an N95 mask if you must go out
- Keep the rescue inhaler with you and take the daily inhaler without missing doses
- Avoid other smoke at home: incense, mosquito coils, wood or kerosene cooking

At this level of PM2.5, a COPD patient's flare-up risk over the following week is about **57% higher** than on a clean-air day (published effect).

⚠️ **Patna: air expected to be very poor on Wednesday, Thursday, Friday (PM2.5 up to 202 µg/m³).**

- Stay indoors as much as possible and keep windows closed
- Wear an N95 mask if you must go out
- Keep the rescue inhaler with you and take the daily inhaler without missing doses
- Avoid other smoke at home: incense, mosquito coils, wood or kerosene cooking

At this level of PM2.5, a COPD patient's flare-up risk over the following week is about **49% higher** than on a clean-air day (published effect).

⚠️ **Delhi: air expected to be very poor on Wednesday, Thursday, Friday (PM2.5 up to 182 µg/m³).**

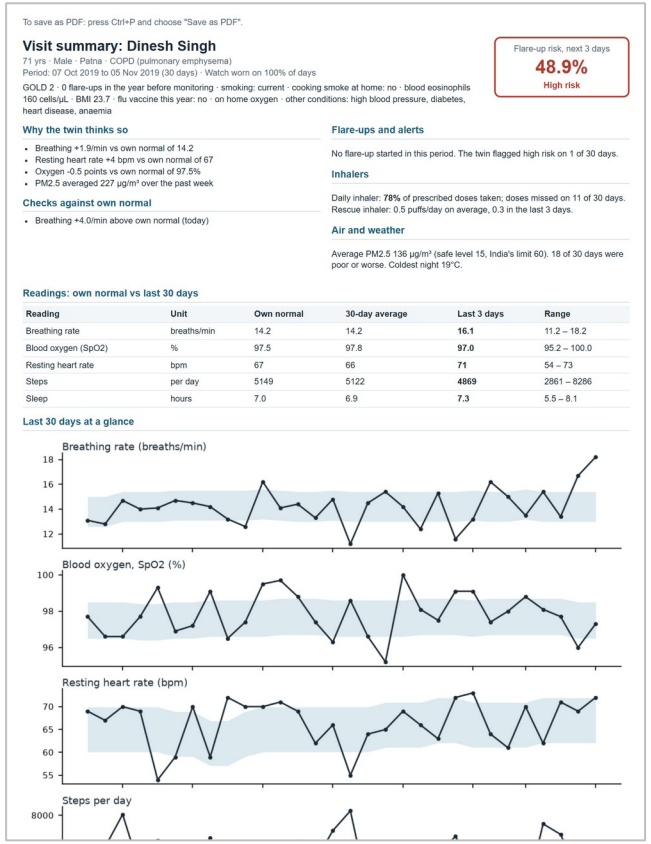
- Stay indoors as much as possible and keep windows closed
- Wear an N95 mask if you must go out
- Keep the rescue inhaler with you and take the daily inhaler without missing doses

⚠️ **Chennai: air expected to be very poor on Wednesday, Thursday, Friday (PM2.5 up to 143 µg/m³).**

- Stay indoors as much as possible and keep windows closed
- Wear an N95 mask if you must go out
- Keep the rescue inhaler with you and take the daily inhaler without missing doses

- Uses India's official **CPCB PM2.5 bands**; advice gets stronger as air gets worse
- Also warns about **cold nights**; shows the extra flare-up risk from the published effect
- *Demo replays real 2019-20 data; a live version would use an air-quality forecast*

## One-page Visit Summary



To save as PDF: press Ctrl+P and choose "Save as PDF".

**Visit summary: Dinesh Singh**

71 yrs · Male · Patna · COPD (pulmonary emphysema)
Period: 07 Oct 2019 to 05 Nov 2019 (30 days) · Watch worn on 100% of days

Flare-up risk, next 3 days
**48.9%**
High risk

GOLD 2 · 0 flare-ups in the year before monitoring · smoking: current · cooking smoke at home: no · blood eosinophils 160 cells/µL · BMI 23.7 · flu vaccine this year: no · on home oxygen · other conditions: high blood pressure, diabetes, heart disease, anaemia

**Why the twin thinks so**

- Breathing +1.9/min vs own normal of 14.2
- Resting heart rate +4 bpm vs own normal of 67
- Oxygen -0.5 points vs own normal of 97.5%
- PM2.5 averaged 227 µg/m³ over the past week

**Checks against own normal**

- Breathing +4.0/min above own normal (today)

**Flare-ups and alerts**

No flare-up started in this period. The twin flagged high risk on 1 of 30 days.

**Inhalers**

Daily inhaler: **78%** of prescribed doses taken; doses missed on 11 of 30 days. Rescue inhaler: 0.5 puffs/day on average, 0.3 in the last 3 days.

**Air and weather**

Average PM2.5 136 µg/m³ (safe level 15, India's limit 60). 18 of 30 days were poor or worse. Coldest night 19°C.

**Readings: own normal vs last 30 days**

| Reading | Unit | Own normal | 30-day average | Last 3 days | Range |
|---|---|---|---|---|---|
| Breathing rate | breaths/min | 14.2 | 14.2 | **16.1** | 11.2 – 18.2 |
| Blood oxygen (SpO2) | % | 97.5 | 97.8 | **97.0** | 95.2 – 100.0 |
| Resting heart rate | bpm | 67 | 66 | **71** | 54 – 73 |
| Steps | per day | 5149 | 5122 | **4869** | 2861 – 8286 |
| Sleep | hours | 7.0 | 6.9 | **7.3** | 5.5 – 8.1 |

**Last 30 days at a glance**

Breathing rate (breaths/min)

Blood oxygen, SpO2 (%)

Resting heart rate (bpm)

Steps per day

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# Who it helps: doctors, health workers and families

## Value by Stakeholder

| Who | Today | With BreatheTwin |
|---|---|---|
| **Doctor / clinic** | Sees the patient only when they come in sick | A ranked daily list of who to call, with reasons |
| **Community health worker** | Visits by rota, not by need | Visits the few patients at highest risk first |
| **Patient and family** | Notice a flare-up when breathing is already bad | Early warning, smog advice, a summary to take to the doctor |
| **Health system** | Emergency visits in smog season | Aim: fewer emergency visits through early action |

## Why it fits India

- Built for **Indian triggers**: cooking smoke, bidis, winter smog
- Covers **ages 22 to 92**, since COPD starts earlier in India
- **Plain language** on every page, with a glossary
- **Open source** and free to adapt
- Planned: low-cost pulse oximeter + phone, Hindi

## The care loop

**Watch daily** → **Warn early** → **Act at home** → **Visit with summary** → **Learn and refine**

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# Honest limits, and the road from prototype to real patients

## Limitations and How We Handle Them

| Limitation | How we handle it |
|---|---|
| Patients, readings and flare-ups are simulated | Every effect sourced or labelled ASSUMPTION; real air and weather; stress-tested on 100 real patients |
| No India-specific PM2.5 effect for COPD | Use the global published figure; flag it clearly |
| Synthetic records differ from real ones | Checked against real MIMIC-IV patients; run on 100 real MIMIC-III patients; gaps reported |
| Demo replays past air data | Same rules plug into a live forecast |
| Needs a smartwatch or oximeter | Minimum kit is a watch + oximeter; no-watch kit uses typed readings |

## Roadmap

**Phase 1: Validate**

Partner hospital · test on real Indian COPD records · tune alert line

**Phase 2: Patient app**

"My lungs today" · daily symptom check-in · SMS / WhatsApp alerts · Hindi

**Phase 3: Scale**

Live air-quality forecasts · link to ABDM health records · health-worker rollout

**Next step:**

*Validate on real Indian COPD patients with a hospital partner before any clinical use.*

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# Thank You!

Anushka Tomar (Team leader) · Yash Saxena

*Team MOMENTUM · SDA Bocconi Asia Center*

